



CUBAN NATIONAL CHEMISTRY CONTEST 2019

"Imagination is more important than knowledge."

Albert Einstein

Name(s): _____ Last name(s): _____

School: _____

Municipality: _____ Province: _____

CODE: _____ Test: **DAY 2. TENTH GRADE EXAM**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 2 problem set, with an individual exam for 10th grade. The exam has 3 problems (problem 4, 4 items; problem 5, 10 items; problem 6, 7 items) and a total of 5 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), and each item scoring (in marks) is indicated.

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Equation of ideal gas	$p \cdot V = n \cdot R \cdot T$
Gases constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta E + W$
Zero on Celsius scale	273,15 K	Gibbs' free energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF ELEMENTS

1 1A																		18 8A	
1 H 1.008	2 2A																2 He 4.003		
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18		
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95		
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80		
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3		
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)		
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)		114 Uuq (277)		116 Uuh (277)		118 Uuo (277)		
58 Ce 140.1		59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0					
90 Th 232.0		91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)					

Problem 4. Stoichiometry II (15 points)

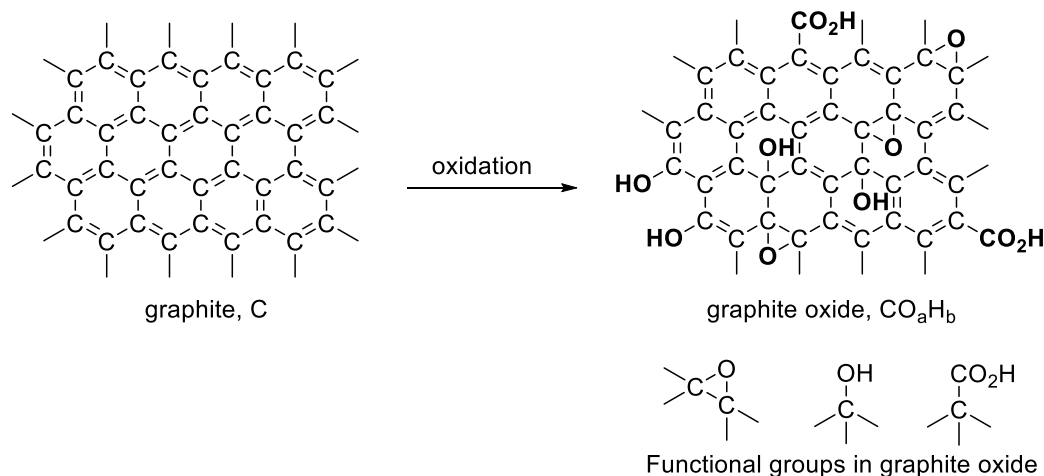
4.1	4.2	4.3	4.4	Total	Total
14	3	3	10	30	15

Compound derivatives from carbon allotropes fullerene and graphite catch the interest of researchers in the field of material sciences.

Buckminsterfullerene (C_{60} , the most abundant fullerene) can form salts of general formula M_nC_{60} ; where M is a metal and n is an integer number. Interest on such compounds was dramatically increased when it was discovered that many are good electric conductors in the solid state. For example, certain alkaline metal **X** combines with C_{60} in different proportions to form salts **A** and **B**, containing 14,0% and 24,6% (in mass) of metal, respectively. **A** is a conductor in the solid state and superconductor below 19,3 K, while **B** is an insulator.

4.1. Find metal **X** and the formula of **A** and **B**.

Graphite oxide (GO) is obtained by treating graphite with strong oxidant reagents. In GO the hexagonal layers of carbon atoms contain oxygen atoms in the form of a variety of functional groups. In recent years, GO has attracted the attention as a plausible precursor of graphene, a carbon two-dimensional nanomaterial with unique electric properties. The molecular formula of GO is CO_aH_b , and the subscripts **a** and **b** depend on the oxidation method.



The simplest model for GO layers, Hoffman's model, assumes that only one functional group ($-O-$) is formed in the carbon layer resulting from graphite oxidation.

4.2. Calculate **a** in the formula CO_a of a GO sample in which 25% of carbon atoms are oxidized (meaning they are attached to an oxygen atom). Assume that Hoffman's model is valid.

4.3. Which is the maximum value that can take **a** in Hoffman's Model?

A more updated model of a GO layer, Lerf-Klinowski's model, considers that the oxygen atoms are present in three functional groups ($-O-$, $-OH$, $-COOH$; see the figure).

4.4. Calculate the amount of carbon atoms (in %) that are **not** oxidized in a layer of GO with formula $CO_{0.42}H_{0.20}$. Suppose that there is an equal amount of $-OH$ and $-COOH$ groups.

Problem 5. Structure and bonding (20 points)

5.1	5.2	5.3	5.4	5.5	5.6	5.7	5.8	5.9	5.10	Total	Total
5	3	2	2	2	3	6	7	4	6	40	20

Boron is present in nature in minerals borax, $\text{Na}_2\text{B}_4\text{O}_5(\text{OH})_4 \cdot 8\text{H}_2\text{O}$ and kernite, $\text{Na}_2\text{B}_4\text{O}_5(\text{OH})_4 \cdot 2\text{H}_2\text{O}$, from which the impure chemical element is obtained. Borax is converted in boric acid by reaction with hydrochloric acid and following dehydration originates boron oxide. The oxide is reduced with magnesium, washed with alkali and then with hydrofluoric acid. In contrast, pure boron is produced by reduction of boron tribromide with hydrogen.

5.1. Formulate the chemical equations corresponding to the reactions previously described.

Boron tends to form four bonds in the compounds containing this atom. This explains that the simplest boron hydride, diborane (B_2H_6), exist in a dimeric form.

5.2. Represent the structure of diborane molecule.

5.3. The reaction between diborane and ammonia generates an ionic product in which only the cation contains nitrogen atoms and both ions contain boron atoms. Suggest a formula for this compound.

The reaction between diborane and lithium hydride forms lithium tetrahydrideborate, LiBH_4 , compound used to storage hydrogen.

5.4. Write the chemical equation for the reaction between LiBH_4 and water to release hydrogen. In this process $\text{Li}[\text{B}(\text{OH})_4]$ is formed.

5.5. Boron halides react with molecules with non-bonding electron pairs, as amines, to form adducts. Why boron halides are electron deficient molecules?

5.6. The reactivity order of boron halides with amines is $\text{BF}_3 < \text{BCl}_3 \leq \text{BBr}_3$. Justify this order considering the molecular structure.

5.7. Write the electronic distribution of molecule BCl_3 according to the Valence Bond Theory. Consider that all the chemical bonds are equivalent.

5.8. BCl_3 molecule decomposes by electronic impact. Use the Molecular Orbital Theory to predict the magnetic properties of product, BCl . Consider only the orbitals from the valence shell.

5.9. When boron trichloride reacts with ammonium chloride borazine, $\text{B}_3\text{N}_3\text{H}_6$, is formed. This compound is analogous to benzene. Borazine molecule is a cycle in which boron and nitrogen atoms are alternating and have sp^2 hybridization. Represent the structure of borazine.

5.10. Boric acid in aqueous solution form a variety of oxyanions depending on the acidity of the media; for example, ions $[\text{B}(\text{OH})_4]^-$ and $[\text{B}_3\text{O}_3(\text{OH})_4]^-$. Represent the structure of these ions.

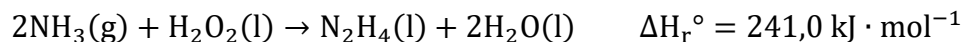
Problem 6. Thermochemistry (15 points)

6.1	6.2	6.3	6.4	6.5	6.6	6.7	Total	Total
2	5	2	2	7	8	4	30	15

Hydrazine, N_2H_4 , is a fuel used in the thrusters of space crafts. It is very popular because doesn't release carbon dioxide. The fast production of hot gasses during the combustion is the source of the rocket trust.

6.1. Write the equation of the decomposition reaction of hydrazine in ammonia and nitrogen, which is a collateral reaction happening in the process.

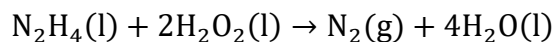
Hydrazine can be obtained by a reaction between ammonia and hydrogen peroxide.



6.2. When hydrazine passes through and adequate catalyst, it decomposes in the parent elements. Calculate the standard enthalpy variation for this process.

$\Delta H_{\text{f}}^\circ$ ($\text{kJ} \cdot \text{mol}^{-1}$): $\text{NH}_3 = -46,1$; $\text{H}_2\text{O}_2 = -187,8$; $\text{H}_2\text{O} = -285,8$.

The first fighter plane thrust by a rocket, Messerschmitt Me 163, was propelled by the reaction of a mixture containing hydrazine and methanol known as "C-Stoff" and hydrogen peroxide ("T-Stoff"). Hydrogen peroxide reacts with hydrazine according to the equation



6.3. What is the oxidation number of nitrogen and oxygen in reactants and products.

6.4. Hydrogen peroxide oxidizes methanol to carbon dioxide and water. Write the chemical equation corresponding to this reaction.

6.5. A fighter plane carries 225 L of hydrazine and 862 L of methanol. Use the following data to calculate the heat released under standard conditions when all the fuel is burnt.

$\Delta H_{\text{combustion}}^\circ$ ($\text{kJ} \cdot \text{mol}^{-1}$): $\text{N}_2\text{H}_4 = -622,2$; $\text{CH}_3\text{OH} = -726,0$;

Density ($\text{g} \cdot \text{mL}^{-1}$): $\text{N}_2\text{H}_4 = 1,021$; $\text{CH}_3\text{OH} = 0,7918$.

Besides obtention, the storage and transport of fuels possess several challenges. In this context, two central concepts are volumetric and gravimetric energy densities. The volumetric energy density is the recoverable energy divided by the total volume. The gravimetric energy density is the recoverable energy divided by the total mass.

6.6. Calculate the volumetric and gravimetric energy densities for hydrazine and methanol.

6.7. Considering the previous result, choose which of the two substances is more adequate to be employed as fuel.